

IS 1008
Introduction to Probability
Distributions

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A discrete r.v. X having pdf of the form

$$P(X=x) = \frac{e^{-\lambda} \lambda^x}{x!} \text{ for } x=0,1,2,3,\dots\text{to infinity}$$

Where λ can take any positive value, is said to follow the Poisson distribution.

If $X \sim \text{Po}(\lambda)$ then it can be shown that $E(X) = \lambda$ and $\text{Var}(X) = \lambda$

Uses of the Poisson Distribution

If an event is randomly scattered in time/space and has mean number of occurrence λ in a given time/space and if X is the r.v. 'the number of occurrences in the given interval, then $X \sim \text{Po}(\lambda)$

Examples.....

Number of

- Car accidents on a particular stretch of a road in one day
- Accidents in a factory in one week
- Telephone calls made to a switch board in a given minute
- Insurance claims made to a company in a given time

Example

(1) If $X \sim \text{Po}(2)$, find

(a) $P(X=4)$

(b) $P(X \geq 3)$

Solution

(1) If $X \sim \text{Po}(2)$, find $P(X=4)$ and $P(X \geq 3)$

Solution

(a) $P(X=4) =$

Solution

(1) If $X \sim \text{Po}(2)$, find $P(X=4)$ and $P(X \geq 3)$

Solution

(b) $P(X \geq 3) =$

Examples

- (1) The mean number of bacteria per milliliter of a liquid is known to be 4. Assuming that the number of bacteria follows a Poisson distribution, find the probability that, in 1ml of liquid, there will be
- (a) No bacteria (b) 4 bacteria (c) less than 3 bacteria

Solution

$X \sim \text{Po}(4),$

(a) No bacteria

(b) 4 bacteria

Solution

$X \sim \text{Po}(4),$

(c) less than 3 bacteria

Example

(2). Using the data of previous exercise find the probability that in 3ml of liquid there will be less than 2 bacteria.

Solution

(2). Using the data of previous exercise find the probability that in 3ml of liquid there will be less than 2 bacteria.

Let Y be the number of bacteria in 3ml of liquid

$$Y \sim \text{Po}(12)$$

$$P(Y < 2) =$$

The Normal Approximation to the Poisson Distribution

If $X \sim \text{Po}(\lambda)$ then $E(X) = \lambda$ and $\text{Var}(X) = \lambda$.

Now for large λ , $X \sim N(\lambda, \lambda)$ approximately.

Example

A radioactive disintegration gives counts that follow a Poisson distribution with mean count per second of 25. Find the probability that in 1 second the count is between 23 and 27 inclusive,

(a) using Poisson distribution

(b) using the Normal approximation to the Poisson distribution

Solution

(a) using Poisson distribution

$$X \sim \text{Po}(25)$$

Solution

(b) using the Normal approximation to the Poisson distribution

using the normal approximation to the Poisson distribution.